

F

FLUTTER COURSE

Session 1 Assignment

Flutter Basics & Dart Fundamentals

Instructor

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Total Marks

100 (+5 Bonus)

Instructions

- Write all answers yourself.
- Do not copy from classmates.
- Use Dart language.
- Write clean and readable code.
- Try every question before asking for help.
- Submit both your Dart code and screenshots of the output.
- Total Marks: 100 (+5 Bonus)

Complete every section in order. Sections progress from easy to more challenging. Blank spaces are provided for handwritten answers and code — use extra paper if needed.

1. What is Mobile Development?

is the process of creating a applications that run on mobile devices and any devices

2. What is Native Development?

code that works only its own platform ex: ios & android

3. What is Cross Platform?

one code that can run on multiple platforms ios & android

4. Mention three Cross Platform frameworks.

flutter, react native, net maui and xamarin

5. What are the native languages for Android and iOS?

Android: Kotlin & iOS: Swift

6. Explain one advantage of Flutter.

write one codebase that can run on multiple platforms.

7. Explain one difference between Flutter and React Native.

flutter: uses dart and same UI in all platforms

React Native : use JavaScript and Native UI components

Write the correct Dart data type for each value.

1.

5

Answer: int

2.

3.14

Answer: double

3.

"Kareem"

Answer: string

4.

true

Answer: boolean

5.

["Ahmed", "Ali"]

Answer: list

6.

```
{  
  "name": "Ahmed"  
}
```

Answer: map

Write Dart code for each of the following.

1. Create an integer variable called age.

Write your Dart code here

2. Create a String variable called studentName.

Write your Dart code here

3. Create a bool variable called isStudent.

Write your Dart code here

4. Create a double variable called salary.

Write your Dart code here

5. Create a List of three names.

Write your Dart code here

6. Create a Map containing: *name*, *age*, *city*

Write your Dart code here

1. What is the difference between var and dynamic?

var: variable's type cannot change afterward.

dynamic: variable can hold values of different types, and Dart checks its type at runtime.

2. When should you use dynamic?

when don't know the type of a value

3. Write one example using var.

Write your Dart code here

4. Write one example using dynamic.

Write your Dart code here

1. What is the difference between final and const?

final: only once and its value is determined at runtime

const: a compile-time constant and cannot be changed.

2. When is final evaluated?

runtime

3. When is const evaluated?

compile time

4. Write one example using final.

Write your Dart code here

5. Write one example using const.

Write your Dart code here

Write Dart code only for each instruction below.

1. Create a List of five cities.

Write your Dart code here

2. Add one city.

Write your Dart code here

3. Remove the third city.

Write your Dart code here

4. Insert a city at index 2.

Write your Dart code here

5. Print the first item.

Write your Dart code here

6. Print the last item.

Write your Dart code here

7. Print the length of the List.

Write your Dart code here

8. Check if the List contains "Cairo".

Write your Dart code here

9. Sort the List.

Write your Dart code here

10. Clear the List.

Write your Dart code here

Write Dart code using each of the following functions.

1. Use map()

Write your Dart code here

2. Use where()

Write your Dart code here

3. Use any()

Write your Dart code here

4. Use every()

Write your Dart code here

5. Use forEach()

Write your Dart code here

1. Create a Student Map containing: *name, age, grade*

Write your Dart code here

2. Print the student's name.

Write your Dart code here

3. Update the age.

Write your Dart code here

4. Add a new key called address.

Write your Dart code here

5. Remove grade.

Write your Dart code here

9

Runtime vs Compile Time

5 Marks

1. What is Compile Time?

before the app runs

2. What is Runtime?

when the app run

B

Coding Challenge (Bonus)

5 Marks

Without looking at your notes, write a complete Dart program that demonstrates ALL of the following:

- | | | |
|-------------------|----------|-------------------|
| ✓ Variables | ✓ int | ✓ double |
| ✓ bool | ✓ String | ✓ List |
| ✓ Map | ✓ var | ✓ dynamic |
| ✓ final | ✓ const | ✓ List operations |
| ✓ Printing output | | |

Write your code below.

Write your complete Dart program here

Challenge Question (Bonus)

Bonus Marks

Create a Student Management program that:

- Stores three students inside a List.
- Each student is represented as a Map.
- Print every student's name.
- Add a new student.
- Remove one student.
- Print the final List.

(Write the complete code.)

Write your complete Dart program here